ABM BUSINESS LOGIC BIBLE

§7 — AI PERSONALIZATION ENGINE (Module 5, Multi-Agent)

V2 Section 7 · Where Intelligence Becomes Persuasion · 7 Agents · Grounded-Only Generation

§6 produced the dossier. §7 turns it into a message. This is where every architectural rule we've built up to now converges: M0 hallucination governance, EPIS-RCM grounding, API-09 ungrounded-personalization rejection, §6 v2's two-layer initiative input (Public + Inferred), and §6 v3's Residual Uncertainty Register as a grounding gate. V2 specifies a seven-agent multi-agent architecture; this section transcribes it operationally and bridges every agent to its governing rules.

Agents	7 (V2)	**5-Whys chains**	5 (root-caused)
New rules	38	**Open questions**	2 flagged

WHY §7 IS WHERE EVERY FAILURE MODE CONVERGES

§7 is the only module that **writes outbound**. Every prior module observed, scored, classified, allocated, enriched. §7 commits something irreversible: a message to a real person at a real bank. The failure modes are correspondingly larger — hallucinated personalization addresses someone wrong; an ungrounded claim damages credibility; a confidently-stated inference about internal politics is reputationally harmful. §7 inherits the strictest gates the architecture has built: grounded-fact contract, RCM stamps, Residual Uncertainty Register, hallucination governance, and a final QC layer (§8) before any send.

7.0 WHY MULTI-AGENT — THE ARCHITECTURAL CHOICE

V2 names five failure modes of single-prompt AI: shallow reasoning, generic outputs, hallucination risk, no quality gates, and inability to specialize. The multi-agent architecture exists because each agent specializes in one reasoning operation, each runs at the temperature appropriate to that operation, each is gated against its predecessor, and a failure at any stage is detectable rather than absorbed into a final output. The output is not 'a personalized message' — it is a **chain of reasoning** whose final artifact happens to be a message.

#	Agent	Reasoning operation	Temperature	Reads from
1	Signal Analyst	What this signal means strategically	0.3 (factual)	Raw signal + normalized summary
2	Account Research	What this organization is doing and why	0.3 (factual)	§6 dossier + AI summary
3	Persona Psychology	What motivates this role	0.3 (factual)	Contact title/seniority/persona_type
4	Pain Inference	Operational pains this persona faces NOW	0.3 (factual)	Outputs of 1+2+3
5	Strategy	Angle, proposition, CTA, tone	0.5 (some creativity)	Outputs of 1–4
6	Message Generation	The actual outreach text	0.4 (creative + controlled)	Strategy brief + all prior
7	Quality Control	APPROVED / REJECTED with reasons	0.1 (strict)	Generated message + chain

THE CHAIN-OF-REASONING DISCIPLINE

No agent runs on its own input alone; each reads the prior agents' outputs as grounding. A failure at Agent 3 (persona psychology) cannot be papered over at Agent 6 (message generation) — the chain breaks at the failure point. This is also what makes §8 (QC) tractable: it has the entire reasoning trace to audit, not just the final message. Per V2 Principle 5: if any step is skipped, the output is REJECTED. Partial reasoning = rejected output.

5-WHYS — WHY THIS PERSONALIZATION ARCHITECTURE

Why #1: Why seven agents instead of one well-written prompt?

Because a single prompt that must research the account AND analyze the persona AND infer the pain AND choose the strategy AND write the message AND validate it cannot be expert at any of them. The reasoning becomes shallow precisely because the context window is asked to do too much. Multi-agent decomposes the cognitive load: each agent is small enough to be expert, each output is small enough to be auditable, and the chain between them is the spec's enforcement surface.

Why #2: Why does temperature vary per agent (0.3 → 0.5 → 0.4 → 0.1) rather than uniform?

Because each agent's job has a different optimal randomness. Research (Agents 1–4) requires factual grounding — low temperature, deterministic. Strategy (Agent 5) requires some creativity to find a non-obvious angle. Message generation (Agent 6) requires creative-but-bounded output. QC (Agent 7) must be near-deterministic — a QC agent that is 'creative' is broken. The temperature curve encodes that the personalization process is mostly inference with one narrow band of generation in the middle.

Why #3: Why must every personalization claim trace to a sourced fact (M0 + API-09 hallucination gate)?

Because the cost of a wrong claim in enterprise outreach is asymmetric: a 'personalized' message that gets the bank's strategy wrong damages credibility worse than no personalization at all. The grounded-fact contract (API-09) rejects any output that contains a personalization claim not traceable to a stored fact — turning hallucination from a probabilistic failure into a deterministic rejection. The Residual Uncertainty Register (§6 v3 ENR-BUDGET-03) binds further: items the engine has decided it cannot know cannot be used as personalization grounds.

Why #4: Why does Agent 5 (Strategy) commit to ONE primary angle rather than offering options?

Because a buffet of angles is not strategy — it is unwillingness to decide. Real messaging requires a single decisive frame; a message that hedges between three value propositions reads as templated by every persona it addresses. V2's constraint is explicit: 'Must select ONE primary angle. Not a buffet of options.' This is also what lets §8 QC audit the message — there is one strategy to validate against, not three.

Why #5: Why does every message require human review during MVP, even ones that pass QC?

Because the engine is operating in an enterprise B2B context where reputation damage compounds: a single embarrassingly-wrong outreach to a Saudi bank executive can poison the relationship for years. Human review during MVP is the safety net that lets the engine ship with confidence; the post-MVP relaxation rule (≥ 85% QC approval rate AND <10% human override) requires empirical evidence that the AI's calibration is good enough — not just a quarterly milestone. P1/P2 signals retain human review permanently because they are the highest-stakes outputs.

Root cause: personalization must be a chain of specialized reasoning, temperature-tuned per operation, grounded-fact-gated end-to-end, single-angle-committed, and human-reviewed during MVP — because writing outbound is the engine's only irreversible action and its failure modes are the most expensive.

AGENT 1 — SIGNAL ANALYST (temperature 0.3)

Reads the triggering signal. Asks: **what does this mean for this account's technology trajectory?** Why is this signal significant, what change does it represent, what downstream effects will it have? Output: 100–200 word strategic interpretation with stated confidence level.

Rule ID	Specification
AI-AG1-01	Agent 1 input: raw_content + normalized_summary + signal_type + signal_strength + source + RCM stamp (from §3 INT-SIG).
AI-AG1-02	Agent 1 output: 100–200 word strategic interpretation answering: what changed, why it matters strategically, what likely downstream effects follow. Output carries a stated confidence level (HIGH / MEDIUM / LOW) derived from the input signal's confidence_score.
AI-AG1-03	Agent 1 must use ONLY facts present in the signal record. It may not invent details about the broader account context — that is Agent 2's job. Cross-agent fact-borrowing is rejected by QC (Agent 7).
AI-AG1-04	Agent 1 output that contradicts the signal's stated facts (e.g. interprets a leadership departure as a hiring event) is REJECTED — the chain breaks here, not at message generation.

AGENT 2 — ACCOUNT RESEARCH (temperature 0.3)

Reads the §6 dossier and the AI account summary. Builds a comprehensive understanding of the organization: what they're trying to achieve, where they are in their digital journey, competitive position. Output: 300–500 word account intelligence brief, every claim cited.

Rule ID	Specification
AI-AG2-01	Agent 2 reads §6 enrichment dossier, the 500–800 word AI account summary (ENR-SUM-01), the SIG-SUBS subsidiary intelligence, the SIG-VENDOR registry, and the SIG-REG regulatory landscape for this account.
AI-AG2-02	Agent 2 output: 300–500 word account intelligence brief. Every claim cites its source (signal, enrichment fact, dossier entry); uncited claims are REJECTED.
AI-AG2-03	Agent 2 must flag knowledge gaps explicitly. Items present in the account's Residual Uncertainty Register (§6 v3 ENR-BUDGET-03) appear in the brief as named unknowns; they are NOT inferred or asserted as facts.
AI-AG2-04	Agent 2 reads the TWO initiative layers from §6 v2 ENR-INIT-DEEP-01: Public Strategic Initiatives (sourced, high-confidence, content layer) and Inferred Buying Drivers (causal inference, RCM-stamped, angle-gating layer). Both are present in the brief; their confidence asymmetry is preserved.
AI-AG2-05	Agent 2 may not import facts about other accounts even from Shared Intelligence Assets (§6 v3 ENR-PORT-03) unless those facts are explicitly tagged as ecosystem-level reusable. Per-account vendor exclusivity (Backbase = SNB only, Efigence = Riyad only) binds: SIG-VENDOR-01.

AGENT 3 — PERSONA PSYCHOLOGY (temperature 0.3)

Analyzes what motivates the target contact's role. KPIs, what keeps them up at night, decision criteria, communication preferences, likely objections. Output: 150–250 word persona profile based on **role analysis, not personal assumptions**.

Rule ID	Specification
AI-AG3-01	Agent 3 input: contact's title, seniority_level, persona_type, department, language preference (EN/AR).
AI-AG3-02	Agent 3 output: 150–250 word persona profile covering primary motivations, decision criteria, communication preferences, likely objections. Output is role-based reasoning ('a CTO at a Saudi bank typically cares about …'), never personal psychology ('this specific person feels …').
AI-AG3-03	Agent 3 is bounded by EPIS-ETH-01: the engine does NOT infer private psychology of a named individual. Personal traits, emotional states, political stances, family or religious context — all forbidden. Persona analysis operates strictly at role level.
AI-AG3-04	Agent 3 uses persona_type taxonomy from V2 §2.2: CTO / CXO / VP_ENGINEERING / HEAD_DIGITAL / PRODUCT_MANAGER / COMPLIANCE / PROCUREMENT / C_SUITE. Each persona type has a documented motivation/concern profile; Agent 3 specializes the profile to this contact's specific role context.
AI-AG3-05	For KSA banking context, Agent 3 specializes: a CTO at an Islamic bank has different priorities than a CTO at a digital-only neo-bank; Agent 3 reads the account's segment (V2 §2.1) and incorporates segment-specific role pressures.

AGENT 4 — PAIN INFERENCE (temperature 0.3)

Reads outputs of Agents 1, 2, 3. Infers the 2–3 most likely operational pains this persona is facing RIGHT NOW given the signal context and account situation. Output: ranked list with confidence and reasoning for each.

Rule ID	Specification
AI-AG4-01	Agent 4 input: Agent 1 strategic interpretation + Agent 2 account brief + Agent 3 persona profile.
AI-AG4-02	Agent 4 output: 2–3 ranked pains, each with: (a) named pain, (b) reasoning chain from signal+account+persona to this specific pain, (c) confidence level, (d) Decimal capability that addresses it.
AI-AG4-03	Pains must be INFERRED from data, not assumed generically. V2 example: 'Probably has manual processes' is too generic and REJECTED. 'Their 2/5 App Store rating combined with the new CTO's mandate suggests onboarding friction that a no-code origination layer could fix' is properly inferred and accepted.
AI-AG4-04	If a candidate pain depends on a fact present in the Residual Uncertainty Register (§6 v3 ENR-BUDGET-03), the pain is downgraded or removed. The engine cannot infer pain from things it has declared unknowable.
AI-AG4-05	Pain inference is governed by the Inferred Buying Drivers layer from §6 v2 ENR-INIT-DEEP-01: if the inferred pain conflicts with the Inferred Buying Driver for the target persona (e.g. pitching cloud to a CTO whose SIG-TENSION posture is pro-on-prem), the pain is downgraded and Agent 5 receives a conflict-flag in its input.

AGENT 5 — STRATEGY (temperature 0.5)

The decisive agent. Reads outputs of 1–4, decides the outreach strategy: what angle do we lead with, what Decimal capability do we highlight, what is the CTA, what tone, what channel.

Rule ID	Specification
AI-AG5-01	Agent 5 output: a single outreach strategy brief containing: lead angle (the ONE topic the message will be about), value proposition (the Decimal capability tied to the inferred pain), proof point (the specific case study, statistic, or comparable to use), call-to-action (low-friction, single ask), tone guidance (formal / consultative / peer-to-peer), channel recommendation (email-first or LinkedIn-first).
AI-AG5-02	Agent 5 must select ONE primary angle. Not two, not three. A strategy that offers options is REJECTED. V2 explicit: 'one clear, decisive strategy.'
AI-AG5-03	Angle selection is gated by §6 v2 ENR-INIT-DEEP-02: the topic comes from the Public Strategic Initiatives layer; the angle (how to pitch the topic) is validated against the Inferred Buying Drivers layer. A topic that conflicts with the inferred driver for the target persona is downgraded; Agent 5 either selects a safer angle or escalates to a persona-generic angle per ENR-INIT-DEEP-03.
AI-AG5-04	When Inferred Buying Driver confidence is below the EPIS-RCM-04 threshold, Agent 5 defaults to safer persona-generic angles (compliance, scalability, operational efficiency) rather than contested persona-specific angles. Anti-false-confidence applied to message strategy.
AI-AG5-05	Agent 5 reads ACCESS_STRENGTH for the resolved SIG-PATH: a P1 path with high access strength permits more direct CTAs ('15-minute call?'); a cold P5 path requires lower-friction CTAs ('worth a 2-minute video?'). The tone/CTA scales with path warmth.
AI-AG5-06	Channel recommendation respects the contact's verified-email status (§6 ENR-EV-01): RISKY or INVALID email forces LinkedIn-first regardless of strategic preference.

AGENT 6 — MESSAGE GENERATION (temperature 0.4)

The only agent that writes the outbound text. Receives the complete strategy brief plus all prior context. Writes the email (subject + body) or LinkedIn message under V2's hard constraints.

V2's hard constraints — Agent 6 must satisfy ALL:

Rule ID	Specification
AI-AG6-01	Length: email body ≤ 150 words; LinkedIn message ≤ 80 words; email subject line ≤ 60 characters. Source: V2 §7.2 Agent 6.
AI-AG6-02	Must reference the specific triggering signal or strategic context — no generic openers.
AI-AG6-03	Must include exactly ONE specific Decimal capability or proof point — not two, not none.
AI-AG6-04	Must end with a low-friction CTA. Forbidden CTAs: 'buy our product', 'jump on a call' (without time-box), any open-ended ask. Permitted: '15-minute call?', '2-minute Loom?', 'worth comparing notes?'.
AI-AG6-05	Banned phrases (V2 explicit): 'I hope this email finds you well', 'Just reaching out', 'I wanted to touch base', 'Synergy', 'Leverage', 'Circle back'. §8 QC enforces hard-rejects.
AI-AG6-06	No false familiarity: 'As a fellow [X]', 'As someone who also …' may NOT appear unless relationship_status ∊ {WARM, ACTIVE, CUSTOMER}.
AI-AG6-07	Seniority-matched framing: C-suite gets strategic framing (vision, market shift, capital efficiency); Directors get operational framing (KPIs, team productivity, time-to-deliver); Managers get tactical framing (specific tool, specific problem, specific saving).
AI-AG6-08	Every personalization claim in the message must trace to a fact in Agent 2's brief (which itself traces to §6 dossier facts). Agent 6 may NOT introduce new facts that did not appear in the chain. Unsourced claims are REJECTED by API-09 hallucination gate before the message ever reaches QC.
AI-AG6-09	Facts present in the Residual Uncertainty Register (§6 v3 ENR-BUDGET-03) may NOT be referenced as known facts in the message. They may be acknowledged as questions ('curious how you're thinking about budget for X this year') but never asserted ('I know your budget is constrained …').
AI-AG6-10	Vendor exclusivity binds: a fact about Backbase at SNB cannot be referenced in a message to Riyad Bank; a fact about Efigence at Riyad cannot be referenced in a message to SNB. SIG-VENDOR-01 propagates into message text via this rule.

AGENT 7 — QUALITY CONTROL (temperature 0.1)

The gatekeeper. Reads the generated message PLUS the entire reasoning chain. Validates against the quality, accuracy, and enterprise-suitability criteria detailed in §8 (the QC module specifies the checks; Agent 7 executes them).

Rule ID	Specification
AI-AG7-01	Agent 7 input: Agent 6's generated message + the complete chain (Agents 1–5 outputs).
AI-AG7-02	Agent 7 output: APPROVED / REJECTED with a specific named reason per the §8 QC check taxonomy (QC-001 hallucination, QC-002 tone, QC-003 length, QC-004 CTA, QC-005 personalization, QC-006 banned phrase, QC-007 compliance).
AI-AG7-03	Agent 7 runs at temperature 0.1 — strict, near-deterministic. A QC agent that varies its verdict on identical input is structurally broken.
AI-AG7-04	REJECTED messages are routed back to Agent 6 for regeneration with the specific failure reason as an additional constraint (V2 QC-REGEN-002). Maximum 3 regeneration attempts (V2 QC-REGEN-001); after 3 failures, the message is flagged MANUAL_CREATION for human authoring.
AI-AG7-05	Agent 7's verdict is the AI layer's final word. APPROVED messages still proceed to MVP-mandatory human review (§8 QC-HUMAN-001); REJECTED messages never reach human review.

7.1 LANGUAGE RULES

Rule ID	Specification
AI-LANG-01	Default outreach language: English. Source: V2 AI-LANG-001.
AI-LANG-02	If contact's preferred_language = AR → generate in Arabic. The Arabic message must be reviewed by an Arabic-fluent human before sending (mandatory, no post-MVP relaxation). Source: V2 AI-LANG-002.
AI-LANG-03	LinkedIn messages match the language of the contact's LinkedIn profile activity (V2 AI-LANG-003).
AI-LANG-04	NEVER mix languages in a single message. Either full English or full Arabic. Source: V2 AI-LANG-004.
AI-LANG-05	Arabic personalization is held to the same grounded-fact contract; Agent 6 generates in the target language but every claim still traces to a stored fact in the dossier.

7.2 MODEL & TEMPERATURE

Rule ID	Specification
AI-MODEL-01	Primary model: Claude Sonnet (latest available version). Fallback: GPT-4o. Source: V2 AI-MODEL-001.
AI-MODEL-02	Temperatures per agent: Research agents (1–4) at 0.3 (factual, grounded); Strategy (5) at 0.5 (some creativity); Message Generation (6) at 0.4 (creative but controlled); QC (7) at 0.1 (strict). Source: V2 AI-MODEL-002.
AI-MODEL-03	If primary model fails (timeout, error), retry ONCE with same model. If second attempt fails, try fallback model. If fallback also fails, queue for manual handling. Source: V2 AI-MODEL-003.
AI-MODEL-04	Model change (e.g. swap primary Sonnet version) is an A4 governance action per GOV-MOD-01; live A/B between models requires explicit governance approval and a defined abort criterion (GOV-MOD-02).
AI-MODEL-05	Every agent invocation is logged with: prompt, output, model, temperature, latency, decision_id (EXEC-05 correlation). Logged to OBS as the AI prompt/output audit surface per OBS-02.

7.3 OPEN-DESIGN-QUESTIONS — §7

Two questions surface in §7 that V2 + the architectural foundation cannot resolve. Both are flagged with candidate options, trade-off, and engineering safe defaults.

OPEN-Q7.1 — POST-MVP RELAXATION CRITERION FOR HUMAN REVIEW

Question: V2 QC-HUMAN-004 specifies the relaxation rule: after 90 days, if AI QC approval rate ≥ 85% AND human override rate < 10%, P3/P4 signals may bypass human review. But these are aggregate statistics — should the relaxation apply per-segment (different threshold per Islamic Bank vs Digital Bank vs BNPL), per-language (EN vs AR), per-persona, or globally?

Candidate options: (a) Global — simplest, may relax review where it shouldn't. (b) Per-segment — each segment crosses the threshold independently; protects against segment-specific failure patterns. (c) Per-language — separate threshold for AR; recognizes that AR review is harder to relax. (d) Per-segment AND per-language.

Trade-off: (a) ships fastest; (d) is safest but requires far more data before any relaxation. (b) and (c) are reasonable middle grounds.

Safe default: Ship (d) per-segment AND per-language. Most conservative; matches the architectural preference for false-confidence avoidance. Each cell relaxes independently when its own data justifies it.

OPEN-Q7.2 — AGENT 5 ANGLE-CONFLICT RESOLUTION DEPTH

Question: When the Public Strategic Initiative and the Inferred Buying Driver for the target persona are in conflict (per §6 v2 ENR-INIT-DEEP-02), AI-AG5-03 says the angle is downgraded. But to what? Three concrete options exist.

Candidate options: (a) Switch to a different Public Initiative topic from the dossier's top 3 (try the next-ranked topic). (b) Switch to a persona-generic safe angle (compliance, scalability, operational efficiency — per AI-AG5-04). (c) Switch to a different target contact at the same account (a different persona may have a non-conflicting inferred driver).

Trade-off: (a) keeps account context, may still hit a different conflict. (b) is always safe but loses the signal-triggered relevance. (c) preserves angle but may pick a less senior contact.

Safe default: Try (a) first, then (b) if a second topic also conflicts. (c) is reserved for major accounts where multi-threaded outreach is explicitly desired — it requires re-running Agents 3–4 for the new contact and is not the default fallback.

§7 RULE ROLL-UP & CROSS-LINKS

Agent / area	Rules	Key cross-links
Agent 1 Signal Analyst	AI-AG1-01..04	§3 INT-SIG signal + RCM stamp, EPIS source/bias
Agent 2 Account Research	AI-AG2-01..05	§6 dossier + ENR-SUM, §6 v3 Residual Uncertainty Register, §6 v2 two-layer initiatives, SIG-VENDOR-01 exclusivity
Agent 3 Persona Psychology	AI-AG3-01..05	V2 §2.2 persona taxonomy, EPIS-ETH-01 (no individual psychology), §1 segment
Agent 4 Pain Inference	AI-AG4-01..05	Residual Uncertainty Register, §6 v2 Inferred Buying Drivers
Agent 5 Strategy	AI-AG5-01..06	§6 v2 ENR-INIT-DEEP-02/03, ACCESS_STRENGTH, §6 ENR-EV verified-email gating
Agent 6 Message Generation	AI-AG6-01..10	API-09 hallucination gate, EPIS-RCM-04 anti-false-confidence, SIG-VENDOR-01 exclusivity, Residual Uncertainty Register
Agent 7 QC	AI-AG7-01..05	§8 QC check taxonomy, V2 QC-REGEN, MVP human-review chain
Language	AI-LANG-01..05	V2 §7.3, grounded contract
Model + temperature	AI-MODEL-01..05	EXEC-05 decision_id, GOV-MOD-01/02, OBS-02 prompt/output audit

§7 COMPLETE — WHAT IT ESTABLISHED

AI Personalization is now operationally specified at V2's level of concretion AND architecturally bridged to every governing rule built in Sections 1–6. The 7-agent multi-agent architecture is transcribed with explicit I/O per agent, per-agent temperature, grounded-fact contract enforcement, the two-layer initiative input from §6 v2, the Residual Uncertainty Register gate from §6 v3, vendor exclusivity propagation into message text, and the V2 hard message constraints (length, banned phrases, no false familiarity, seniority-matched framing). 38 new rules; 2 open questions flagged with safe defaults.

§7 is where the engine's intelligence becomes persuasion — and where the architectural failure modes that appear here (message-strategy hallucination, narrative drift, persona-mismatch, cross-channel inconsistency) are the genuinely new class of problems. §8 (QC) operationalizes the gate that catches them before send.

PLACEMENT MAP UPDATE

§7 status: **Complete-with-Opens** (OPEN-Q7.1 relaxation criterion, OPEN-Q7.2 angle-conflict resolution depth). Engineering may build against the safe defaults; resolutions are A4 governance actions per TIER-OPEN-DISCIPLINE-02.

ON THE HORIZON

§8 Quality Control Engine — the 7 QC checks (QC-001–007: hallucination, tone, length, CTA, personalization, banned phrase, compliance), regeneration limits (≤3), human-review SLA (4h business hours), and the relaxation rule §7's OPEN-Q7.1 binds. §8 is the operational gate that catches §7's failure modes before they reach a prospect.